

Musikalische Akustik 2 SE Project "Proposal"

Physical modelling of one channel of a diatonic harmonica

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Abstract

This proposal presents a reduced physical model of one single channel of a diatonic harmonica. The model combines four main components:

- reed dynamics represented by equivalent damped oscillators
- nonlinear airflow through the reed openings based on a Bernoulli-type reduction
- lumped acoustic descriptions of the channel chamber
- a reduced representation of the vocal tract

The aim is not to reproduce every geometric and aerodynamic detail of the instrument, but to formulate a compact, physically interpretable model that captures the essential coupling between blow and draw reeds, chamber pressure, and vocal-tract loading. The resulting continuous-time system is intended as the basis for a direct numerical simulation in the context of physical modelling.

1 Model Equations

1.1 General single-reed model (reduced free-reed formulation, cf. [1, 3, 10])

Millot et al. [1] support modelling a harmonica reed as an equivalent one-degree-of-freedom damped oscillator. In addition, standard mass-spring(-damper) system descriptions and state-space formulations provide the general dynamical framework for such second-order models [10]. In the present reduced model, the aerodynamic driving force is represented by a lumped pressure-loading term acting on an effective reed surface:

$$m_i x_i'' + r_i x_i' + k_i x_i = F_{\text{air}} \quad (1)$$

$$F_{\text{air}} = S_i(p_m - p_c) \quad (2)$$

Here, equation (2) is understood as the lumped approximation

$$F = \int_S \Delta p \, dS \approx S_i \Delta p,$$

where S_i denotes an effective pressure-loaded reed surface. Thus, the force law is written in terms of pressure acting on an effective surface, while the opening area is reserved for the airflow model below.

1.2 Two-reed channel model

A diatonic harmonica channel contains two coupled reeds, namely a blow reed and a draw reed. In the reduced model, the force on each reed is approximated by the corresponding pressure difference acting on an effective reed surface.

Blow reed (mouth $\rightarrow$ chamber):

$$F_b = S_b(p_m - p_c) \quad (3)$$

Draw reed (chamber $\rightarrow$ outside):

$$F_d = S_d(p_c - p_{\text{out}}) \quad (4)$$

Equations (3) and (4) are therefore to be understood as lumped pressure-loading closures for the blow and draw reed, consistent with the free-reed channel geometry.

1.3 Equivalent Oscillator Parameters (after [1])

$$K_i = \frac{16}{15} K_{\text{cal},i} \quad (5)$$

$$m_i = \frac{K_i}{(2\pi f_{r,i})^2} \quad (6)$$

$$r_i = \frac{K_i}{2\pi f_{r,i}} Q_i^{-1} \quad (7)$$

The stiffness parameter in the reed equation is identified as $k_i = K_i$.

1.4 Nonlinear Airflow (Bernoulli-based reduction, cf. [2, 3])

Following the one-dimensional free-reed description used in the literature, the volume flow at reed j can be decomposed into a gap-flow contribution and an additional term caused by the motion of the reed itself:

$$Q_j \approx (-1)^j W h_j v_j + S_j \dot{h}_j.$$

Here, $W h_j$ represents an effective instantaneous gap area, while $S_j \dot{h}_j$ accounts for the extra volume flow generated by reed motion. Assuming a stationary Bernoulli relation in the gap gives the jet velocity in the reduced form

$$v_j \approx \text{sgn}(\Delta p_j) \sqrt{\frac{2|\Delta p_j|}{\rho}}.$$

In this model, the explicit moving-reed contribution $S_j \dot{h}_j$ is neglected in order to obtain a simpler lumped nonlinear flow law. In addition, geometric details such as contraction and vena-contracta effects are absorbed into a discharge coefficient C_j . With the pressure differences

$$\Delta p_b = p_m - p_c, \quad \Delta p_d = p_c - p_{\text{out}},$$

this leads to the reduced Bernoulli/orifice approximations

$$Q_b = C_b A_b(x_b) \text{sgn}(p_m - p_c) \sqrt{\frac{2|p_m - p_c|}{\rho}} \quad (8)$$

and

$$Q_d = C_d A_d(x_d) \text{sgn}(p_c - p_{\text{out}}) \sqrt{\frac{2|p_c - p_{\text{out}}|}{\rho}}. \quad (9)$$

Thus, equations (8) and (9) are not meant as literal copies of the source equations, but as reduced time-domain airflow closures derived from the Bernoulli-based free-reed formulation.

1.5 Chamber Pressure Dynamics (lumped acoustic compliance, cf. [7, 8, 9])

The chamber pressure is described by a standard lumped acoustic compliance relation for a compressible cavity. This form is consistent with the continuity relation, the compressibility of air, and the identification of ρc^2 as the bulk modulus of the medium:

$$p'_c = \frac{\rho c^2}{V_c}(Q_b - Q_d) \quad (10)$$

1.6 Vocal Tract Impedance (following [4], cf. [8, 9])

$$Z(\omega) = \frac{P(\omega)}{Q(\omega)} \quad (11)$$

1.7 Vocal Tract Resonator Model (reduced time-domain approximation, cf. [4, 5, 8, 9])

To obtain a tract model usable in time-domain simulation, the vocal tract is represented by a reduced one-mode resonator. This is consistent with the lumped acoustic interpretation of compliance, acoustic mass, and acoustic resistance, and with the observation that vocal-tract geometry acts as a variable acoustic load:

$$p''_t + \frac{\omega_t}{Q_t} p'_t + \omega_t^2 p_t = \omega_t^2 Z_t(Q_b - Q_d) \quad (12)$$

1.8 Numerical Implementation Perspective

The reduced continuous-time model given above is intended as the basis for a direct numerical simulation. This follows the general physical-modelling framework described by Bilbao [6], where simple oscillators serve as the starting point and are then discretized for time-domain simulation.

Nomenclature

m_i	effective mass of reed i
r_i	damping coefficient of reed i
k_i	stiffness parameter in the reed ODE
K_i	stiffness derived from calibration
x_i	displacement of reed i
x'_i, x''_i	velocity and acceleration of reed i
$K_{\text{cal},i}$	calibrated stiffness of reed i
$f_{r,i}$	natural frequency of reed i
Q_i	quality factor of reed i
F_{air}	aerodynamic driving force (single-reed model)
F_b, F_d	aerodynamic forces on blow and draw reed
S_i	effective pressure-loaded surface of reed i
S_b, S_d	effective pressure-loaded surfaces of blow and draw reed
$A(x)$	effective opening area (single-reed model)
$A_b(x_b), A_d(x_d)$	opening areas of blow and draw reeds
W	effective reed width in the reduced gap model
h_j	effective gap opening of reed j
$\dot{h}_j$	time derivative of the effective gap opening
S_j	effective coefficient for reed-motion-induced flow
v_j	jet velocity through the reed opening
Δp_j	pressure difference across reed j
Q	volume flow rate
Q_b, Q_d	volume flows through blow and draw reeds
C_b, C_d	discharge coefficients
ρ	air density
p_m	mouth pressure
p_c	chamber pressure
p_t	vocal tract pressure
p_{out}	ambient pressure
V_c	chamber volume
c	speed of sound
ρc^2	bulk modulus of air

$Z(\omega)$	acoustic impedance
$P(\omega)$	pressure (frequency domain)
$Q(\omega)$	flow (frequency domain)
ω	angular frequency
ω_t	tract resonance frequency
Q_t	tract quality factor
Z_t	tract coupling coefficient
$\text{sgn}(\cdot)$	sign function

2 Approach

Given the listed equations above and the following sources, the proposed approach is to construct a *reduced physical model* of a single diatonic harmonica channel that remains simple enough for seminar-scale implementation while preserving the main coupled mechanisms of sound generation. The model is based on four interacting components:

1. reed dynamics
2. nonlinear airflow
3. chamber-pressure evolution
4. vocal-tract loading

First, each reed is represented as an equivalent one-degree-of-freedom damped oscillator. This follows the experimental and modelling work of Millot et al., who show that harmonica reeds can be described by equivalent mass, damping, and stiffness parameters. In the proposed model, the reed motion provides the mechanical state variables, while the aerodynamic excitation is introduced through a lumped pressure-loading term acting on an effective reed surface.

Second, the airflow through the reed openings is described by a reduced Bernoulli-based nonlinear law. Following the free-reed formulation discussed in the literature, the full flow description contains both a gap-flow term and a contribution caused by the motion of the reed itself. In order to obtain a compact seminar model, the explicit moving-reed flow term is neglected and the remaining nonlinear gap flow is written in Bernoulli/orifice form. This choice is consistent with the view that the principal nonlinearity of the system arises from the airflow through the reed openings.

Third, the harmonica chamber is represented as a lumped acoustic compliance. The chamber pressure is therefore treated as a dynamic variable driven by the difference between inflow and outflow. This introduces the acoustic coupling between blow and draw reed and allows the channel to be modelled as more than a purely mechanical reed pair.

Fourth, the player's vocal tract is included in reduced form. Its frequency-domain role is described through acoustic impedance, while in the time domain it is approximated by a single resonant subsystem. This provides a simple way to represent the fact that vocal-tract geometry modifies the acoustic loading of the instrument and influences playing regimes such as pitch bending.

The overall intention is not to reproduce every aerodynamic and anatomical detail of the real instrument, but to formulate a compact and physically interpretable continuous-time model that captures the essential interactions within one harmonica channel. On this basis, the next step is to discretize the system and implement it as a direct numerical simulation, following the general physical-modelling framework described by Bilbao.

References

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